

# Clinical Risk Assessment Report: Patient 95

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 55.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.52x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET), which elevated the patient's absolute risk by +7.1%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)**      **Value: 0.87 [Top 11%] (+7.1%)**  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)**      **Value: -0.38 [Bottom 6%] (+6.3%)**  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Tumor Pixel Correlation (CT) (GLCM\_Correlation.CT)**      **Value: 0.68 [Avg (75th %ile)] (+4.9%)**  
*Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)**      **Value: 0.74 [Avg (72th %ile)] (+4.1%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)**      **Value: 0.9 [Top 18%] (+3.6%)**  
*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)**      **Value: 1.08 [Top 13%] (+3.6%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)**      **Value: -0.23 [Bottom 8%] (+2.9%)**  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

**High Gray-Level Zone Emphasis (CT) (GLZLM\_HGZE.CT)**      **Value: 1.07 [Top 13%] (+2.6%)**  
*Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.*

## 3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

